# Gauß - Manin connection and F-crystals

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Finiteness + $p$-adic cohom. theory (coeff $W_n(\mathbb{F})$, but not $W(\mathbb{F})$)

<u>Prop</u> Suppose $\mathbb{F}$ field of char $p$. $X/\mathbb{F}$ smooth proper scheme. Let $\mathcal{E}$ be a crystal, then $H^i_{crys}(X/W_n(\mathbb{F}), \mathcal{E})$ is a f.g. $W_n(\mathbb{F})$-module.

<u>Pf.</u> Induction on $n$.

If $n=1$, $H^i_{crys}(X/\mathbb{F}, \mathcal{E}) = H^i_{dR}(X, (\mathcal{E}, \nabla))$ – f.g.

Suppose we know for $n-1$, then have an exact sequence of sheaves on $\operatorname{Spec} W_n(\mathbb{F})$:

$$0 \to \overset{\text{killed by } p}{\widetilde{I}} \to \mathcal{O}_{\operatorname{Spec} W_n(\mathbb{F})} \to \mathcal{O}_{\operatorname{Spec} W_{n-1}(\mathbb{F})} \to 0$$

Consider LES in crystalline cohomology:

$$H^i_{crys}(X, I \otimes \mathcal{E}) = H^i_{dR}(X, (I \otimes \mathcal{E}, \dots))$$

<u>Recall</u>. $H^i_{et}(X, \mathbb{Z}_\ell) := \varprojlim_n H^i_{et}(X, \mathbb{Z}/\ell^n\mathbb{Z})$

$X/\mathbb{F}$ smooth proper, could $H^i_{crys}(X, W(\mathbb{F})) = \varprojlim_n H^i_{crys}(X/W_n(\mathbb{F}), \mathcal{O}_X)$

<u>Def</u>. A $p$-adic base is a noetherian PD ring $(A, I, \delta)$ with a sub-PD ideal $\mathfrak{p}$ of $I$ s.t. $A$ is $\mathfrak{p}$-adically complete, separated, and such that $\mathfrak{p}$ contains some prime number $p$.

We let $A_n = A/\mathfrak{p}^{n+1}$, $S_n = \operatorname{Spec} A_n$, $S = S_1$, $\hat{S} = \operatorname{Spf} A$.

**Def.** Suppose $X/A$ is a PD scheme. Define $\mathrm{Crys}(X/\hat{S})$ – a site whose objects are $(U, T, \delta)$, $U \subset X$ open, $T$ – nilpotent thickening of $U$, and $\delta$ is a PD-structure on $T$, and coverings are Zariski coverings.
s.t. $\exists n, p^n \mathcal{O}_T = 0$

$i_n : \mathrm{Crys}(X/A_n) \to \mathrm{Crys}(X/\hat{A})$.

Define $\mathrm{Crys}(X/S_\cdot)$ – objects are pairs $(n, T)$, $n \in \mathbb{N}$, and $T \in \mathrm{Crys}(X/S_n)$

the coverings are $\{(n, T_i)\} \to (n, T)$ if $\{(T_i)\} \to (T)$ in $\mathrm{Crys}(X/S_n)$

$\mathbb{N}$ $\quad 0 \to 1 \to 2 \to 3 \to 4 \to 5 \cdots$

Observe that inverse systems of sets are the same as sheaves on $\mathbb{N}$.

Observe further that $\Gamma(\mathbb{N}, (A_\cdot)) = \varprojlim_n A_n$

Observe further that a sheaf on $\mathrm{Crys}(X/S_\cdot)$ is a sequence of sheaves

$F_n$ on $\mathrm{Crys}(X/S_n)$ with $F_n|_{\mathrm{Crys}(X/S_{n-1})} \to F_{n-1}$.

$j : \mathrm{Crys}(X/S_\cdot) \to \mathrm{Crys}(X/\hat{S})$, $\quad j(n, (U, T, \delta)) = (U, T, \delta)$.

<u>Goal</u> Prove that $H^i(\mathrm{Crys}(X/\hat{S}), \mathcal{E}) \simeq H^i(\mathrm{Crys}(X/S_\cdot), j_* \mathcal{E})$.

<u>Lemma.</u> Let $X$ be a top space, $(F_n)_{n \in \mathbb{N}}$ be an inverse system of ab. sheaves on $X$. Suppose $\exists$ basis $\mathcal{B}$ of $X$ s.t.

a) $\forall q > 0$, $\forall V \in \mathcal{B}$, $H^q(V, F_n) = 0$

b) The inverse system $H^0(V, F_n)$ satisfies (ML)

$(A_n)$ satisfies (ML) $\iff$ $\forall n \in \mathbb{N}, \exists n' > n, \forall n'' > n'$, $\pi_n^{n''}(A_{n''}) = \pi_n^{n'}(A_{n'})$

Then $(\mathcal{F}_n)$ is $\varprojlim$ - acyclic. $R^q \varprojlim (\mathcal{F}_n) = 0$ as sheaves on $X$ $\forall q > 0$.

$\mathbb{N} \times X \xrightarrow{\pi} X$ , $R^q \pi_* (\mathcal{F}_\cdot) = 0$, $q > 0$.

Take $(I_\cdot^\cdot)$ to be an injective resolution of $(\mathcal{F}_\cdot)$ as a sheaf on $\mathbb{N} \times X$.

$\forall q > 0$, $\forall n \in \mathbb{N}$, $I_n^q$ is a flasque sheaf on $X$, and $I_n^q(U) \twoheadrightarrow I_{n-1}^q(U)$ are epi.

Take $V \in \mathcal{B}$, then $\Gamma(V, I_n^\cdot)$ is acyclic in deg $> 0$, $H^0 = H^0(V, \mathcal{F}_n)$.

therefore, by EGA $0_{III}$ 13.2.3, $H^q(\varprojlim \Gamma(V, I_n^\cdot)) = \varprojlim H^q(V, I_n^\cdot) = 0$

$$\overset{\shortparallel}{R^q \pi_* (\mathcal{F}_\cdot)(V)}$$

<u>Lemma</u>. If $\mathcal{F}$ is an ab. sheaf on $\mathrm{Crys}(X/S_\cdot)$,

a) For $(U, T, \delta) \in \mathrm{Cris}(X/S)$, $(j_* \mathcal{F})(U, T, \delta) = \varprojlim_n \mathcal{F}(n, (U, T, \delta))$

b) If each $\mathcal{F}(n, (U, T, \delta))$ is quasi-coherent, and if the maps

$\mathcal{F}(n', (U, T, \delta)) \twoheadrightarrow \mathcal{F}(n, (U, T, \delta))$ are epi, then $\mathcal{F}$ is $j_*$-acyclic.

$\Rightarrow$

$\mathcal{F}(n, (U, T, \delta)) = (j_n^* \mathcal{F})(U, T, \delta)$ $\quad j_n : \mathrm{Crys}(X/S_n) \to \mathrm{Crys}(X/S_\cdot)$

$$\begin{array}{ccc} \mathrm{Crys}(X/S_\cdot) & \xrightarrow{\text{res}} & \mathbb{N} \times T_{\mathrm{Zar}} \\ \downarrow{\scriptstyle j_*} & & \downarrow{\scriptstyle \pi_*} \\ \mathrm{Crys}(X/\hat{S}) & \xrightarrow{\text{res}} & T_{\mathrm{Zar}} \end{array}$$

If $\mathcal{F} \in (X/S_\cdot)_{\mathrm{crys}}$,

$(R^q j_* \mathcal{F})|_T = (R^q \pi_*)(\mathcal{F}|_T)$

$\square$

Th Suppose $X/A_0$ is smooth, qcqs, and $\mathcal{E}$ is a locally free f.g. crystal of $\mathcal{O}_{X/\hat{S}}$ - modules in $(X/\hat{S})_{crys}$.

a) $\exists$ object $\mathbb{D}$ in the derived cat. of inverse systems of $A$-mods s.t. $D_n \simeq R\Gamma(X/S_n, i_n^* \mathcal{E})$ and $\mathbb{R}\varprojlim \mathbb{D} \simeq R\Gamma(X/\hat{S}, \mathcal{E})$

b) For each $n$, $\exists$ natural iso.

$$R\Gamma(X/\hat{S}, \mathcal{E}) \overset{\mathbb{L}}{\underset{A}{\otimes}} A_n \xrightarrow{\sim} R\Gamma(X/S_n, i_n^* \mathcal{E})$$

c) $R\Gamma(X/\hat{S}, \mathcal{E})$ has finite Tor-dim.

d) If $X/A_0$ is proper, then the complex $R\Gamma(X/\hat{S}, \mathcal{E})$ is perfect. Moreover, the inverse system $(H^i(X/S_n, i_n^*\mathcal{E}))_{n \in \mathbb{N}}$ satisfies (ML), and $\exists$ isom.

$$H^i(X/\hat{S}, \mathcal{E}) \simeq \varprojlim H^i(X/S_n, i_n^* \mathcal{E})$$

Def. A sheaf $\mathcal{F}_\cdot$ of $\mathcal{O}_{X/S_\cdot}$ - modules is a crystal if $\mathcal{F}|_{Crys(X/S_n)}$ a crystal and $\mathcal{F}(n, T) \Rightarrow \mathcal{F}(n', T')$ is an isom.

Pf. $D_n \overset{\mathbb{L}}{\underset{A_n}{\otimes}} A_{n-1} \to D_{n-1}$ are q-isoms.

Consider sheaves on $\mathbb{N}$ = inverse systems.

Def. An object $D \in \mathbb{D}(\mathbb{N}, A)$ is quasi-consistent if the natural maps $D_n \overset{\mathbb{L}}{\underset{A_n}{\otimes}} A_{n-1} \xrightarrow{\sim} D_{n-1}$ are all isomorphisms in $\mathbb{D}(A)$.

Properties - If $D \in D^-(\mathbb{N}, A)$ is quasi-consistent, then map $(R\varprojlim D) \overset{\mathbb{L}}{\underset{A}{\otimes}} A_n \xrightarrow{\sim} D_n$ is an isom.

- If $D_0$ has finite Tor-dimension, then any $D_n$ and $R\varprojlim D_n$ have finite Tor-dim.

- If $D$ is quasi-consistent, and $D_0$ is of finite Tor-dim. then $(H^i(D_n))$ satisfies (ML), $H^i(R\varprojlim D) \xrightarrow{\sim} \varprojlim H^i(D)$ are isos.

- If $D$ is quasi-consistent, then $R\varprojlim D$ is perfect iff $D_0$ is.

Cor. If, in addition $A$ is a complete DVR, then $\exists$ exact sequence
$p = I = m$

$$0 \to H^i(X/\hat{S}, \mathcal{E}) \otimes A_n \to H^i(X/S_n, \mathcal{E}_n) \to \mathrm{Tor}_1^A(H^{i+1}(X/\hat{S}, \mathcal{E}), A_n) \to 0$$

Def $H^i_{crys}(X) = \mathbb{H}^i(X/\hat{S}, \mathcal{O}_{X/\hat{S}})$
$X$ proper

Cor $\exists$ exact seq

$$0 \to H^i_{crys}(X) \otimes_A A_0 \to H^i_{dR}(X/S_0) \to \mathrm{Tor}_1^A(H^{i+1}_{crys}(X), A_0) \to 0.$$

If $Y/\hat{S}$ is a smooth lifting of $X$, then there is an isom. $H^i_{crys}(X) \simeq H^i_{dR}(Y/S)$